

Day 19 Problem Set

Geometry Summer Credit | Arc Length, Sector Area, and Radians

Name:	Date:	Period:
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Directions

Write the fraction $\theta/360$ first. Leave exact answers in terms of π unless told otherwise.

Part A: Arc Length

1.	Find the arc length for radius 10 and central angle 72 degrees.
2.	Find the arc length for radius 14 and central angle 90 degrees.
3.	A circle has diameter 18 and central angle 200 degrees. Find the arc length.
4.	Find the arc length for radius 6 and central angle 150 degrees.
5.	A circle has radius 12 and arc length 6π . Find the central angle.
6.	A circle has central angle 120 degrees and arc length 8π . Find the radius.

Part B: Sector Area

7. Find the sector area for radius 8 and central angle 90 degrees.

8. Find the sector area for radius 12 and central angle 60 degrees.

9. A circle has diameter 20 and central angle 144 degrees. Find the sector area.

10. Find the sector area for radius 15 and central angle 216 degrees.

11. A circle has radius 10 and sector area 25π . Find the central angle.

12. A semicircular sector has area 72π . Find the radius.

Day 19 Problem Set

Continued | Mixed Measurement and Radian Readiness

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| 13. | A circle has radius 9 and central angle 80 degrees. Find both arc length and sector area. |
| 14. | A sector is $\frac{3}{8}$ of a circle. Find its central angle and state what fraction of the circumference and area it represents. |
| 15. | Error analysis: A student uses $\theta/180$ instead of $\theta/360$ for arc length. Explain the effect on the answer. |
| 16. | Convert 30 degrees to radians. |
| 17. | Convert 135 degrees to radians. |
| 18. | Convert $5\pi/6$ radians to degrees. |
| 19. | Convert $3\pi/2$ radians to degrees. |
| 20. | Spiral review: A tangent and secant share an external point. The secant has external 4 and whole 25. Find the tangent length. |
| 21. | Challenge: A wheel with radius 14 inches rotates through 225 degrees. How far does a point on its rim travel? Give an exact arc length. |